

Sisters of Code

SIGGRAPH 2021 Retrospective Panel

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1 INTRODUCTION

This international panel focuses on examining the historical traces of contributions produced by women in fields such as information visualization, infographics, computer arts, and digital aesthetics. The panel has three presentations by artists, technologists, and scholars such as essential and often underexplored legacy:

1.1 “Vera Molnar: ICÔNE/2020” by Francesca Franco

This panel presentation is centred on “ICÔNE/2020,” a new commission that the author initiated, produced and curated with computer art pioneer Vera Molnar in 2020–21. This is Molnar’s first ever sculptural work in a career that spans over 80 years. Taking the new commission as a centre point, Franco’s talk explores the process that made this commission possible, bringing together original sketches and documentation material that reveal the complexities behind the making of this project. By reimagining what art could be in an age of immense technological change, Molnar’s radical adoption of technology and her conceptual approach to computing have inspired Franco to initiate a new collaboration that takes computer art to new unexplored levels. This new commission has been awarded two major grants from the Henry Moore Foundation and

the AHRC that will support “ICÔNE/2020,” an exhibition curated by Francesca Franco in 2021.

1.2 “Women in Data Visualisation” by Sara Akhlaq

This talk aims to highlight contributions of women to the field of data visualisation throughout history until present day. It draws attention to how they have strived to bridge the human understanding and comprehension gap by harnessing the power of lines, shapes and arrangement to represent a wide range of issues. Despite being excluded from the very narrative of the field, there are women who have yet managed to further data visualisation through their ingenuity. Visualising the same data they are excluded from, these women belonging to transdisciplinary fields have strived to equip themselves with visualisation tools to represent data in a more engaging and effective fashion. Gender imbalance has hindered women’s success journey throughout history, and while a lot of names have gone unacknowledged, there are some women who have soldiered through the hurdles of underrepresentation to make their name in data visualisation. From Florence Nightingale’s Rose Diagram showing hygiene-related fatalities in hospitals in 1858 to Mona Chalabi’s data drawings that highlight social issues of our time, this talk takes the audience through some prominent women in their unique contributions to the field of data visualisation.

1.3 “The History of Women’s Contributions to SIGGRAPH Revealed: A Glimpse into the SIGGRAPH Online Archives” by Bonnie Mitchell and Jan Searleman

The history of computing is fraught with gender inequalities related to access, education, mentorship and acknowledgement. Although women such as Ada Lovelace, Grace Hopper, Katherine Johnson have finally gained recognition for their efforts, they were relatively obscure in their own time. In the early days of computing, women often worked on menial tasks such as data entry, computing tables of numbers and delivering coffee to the men in the back room. Young women wishing to enter the field of computer science in the 50s through 80s were often advised to learn to type and take dictation, and actively discouraged from studying programming and electronics. At the university level, computer science courses often

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consisted entirely of men, with the occasional persistent woman. Women in digital art also faced similar issues but were among the first to break through the barrier to define a direction for this exciting new art form. Females in the field of computer animation also found it challenging to excel in a world dominated by men and it was rare to find women in positions of power. Organizations such as ACM SIGGRAPH, established in 1974, provided opportunities for women to serve in leadership roles and gain recognition in their respective fields. In December of 2020, Jan Searleman and Bonnie Mitchell began working on an archive of ACM SIGGRAPH's history. During the development of this online archive, patterns started to emerge which revealed a more complex picture. This session will explore the role of women throughout the organization's history.

2 BIOGRAPHIES

Francesca Franco. Francesca Franco is a Venetian-born art historian based in the UK. She is Senior Research Fellow at the University of Exeter and Visiting Professor at Danube University Krems. The central theme of her research is the history of art and technology and the pioneers of computer art. She is CI on the AHRC-funded project "Documenting digital art: re-thinking histories and practices of documentation in the museum and beyond." In 2017 she was curator-in-residence at the Fondazione Bevilacqua La Masa in Venice where she curated *Algorithmic Signs*, an exhibition that explored the history of generative art. She has published extensively on the history of early computer art and its pioneers. Franco is author of *Generative Systems Art* (Routledge 2018) and *The Algorithmic Dimension* (forthcoming, Springer 2021). She is currently working on her new commission with computer art pioneer Vera Molnar and on her next curatorial project, a large-scale exhibition of computer art for the Venice Biennale.

Sara Akhlaq. Sara Akhlaq is a research associate at Urban Complexity Lab at FH Potsdam. Her main area of interest is digital cultural heritage and ways in which it could be represented and disseminated. Her research focus is on handling cultural data of digital collections through structured metadata. She is also working on design conceptions in information visualisation of cultural data, and building bridges among different modalities in multi-sensory cultural data representation.

Bonnie Mitchell. Bonnie Mitchell is a new media artist, online archive developer and a Professor at Bowling Green State University in Digital Arts, in Bowling Green, Ohio, USA. Mitchell is a member of the ISEA International Advisory Committee and ACM SIGGRAPH History and Digital Arts Committee where she focuses on the development of their online archives. Her current creative practice focuses on development of physically immersive data visualization environments that showcase climate change over time. Mitchell's artworks explore spatial and experiential relationships to our physical, social, cultural and psychological environment

through interaction, abstraction and audio. She often collaborates with Elaine Lillios, a composer, to create abstract visual music installations and animations that have been shown in hundreds of venues world-wide. Bonnie Mitchell was also an early net-artist producing numerous large-scope international internet art projects that were shown at ISEA, SIGGRAPH and online starting in 1992, garnered an Honorable Mention from ARS Electronica in 1995. Collectively, Mitchell's presentations and artworks have been included in festivals, exhibitions, conferences, and symposia in 6 continents and at numerous prestigious events including Ars Electronica, ISEA, SIGGRAPH, Punto y Raya, Visual Music Marathon, Melbourne and London International Animation Festivals, AnimaFest Croatia, iDMAA IDEAS Festival, Society for Electro-acoustic Music, Les Somnests du Cinema d'Animation at the Cinémathèque Québécoise, and many more.

Jan Searleman. Jan Searleman taught Computer Science at Clarkson University for 37 years, retired in 2015, and since retirement has been an Adjunct Research Professor at Clarkson. Her research areas are Virtual Environments, Human-Computer Interaction, and Artificial Intelligence. In 1979, Jan, along with colleague James Lynch, established a major in Computer Science. She was instrumental in creating Clarkson's MS and PhD in Computer Science. Jan created and taught a variety of CS courses, including Artificial Intelligence in 1979, and Computer Graphics in 1980 (in the days of a green dot on a black screen). In the 1990s, she created a student lab in Virtual Reality, and introduced a course on Virtual Environments. A senior member of the ACM since 1976, and of SIGGRAPH since 1978, Jan established Clarkson's ACM student chapter in 1980. She also created Clarkson's ACM SIGGRAPH student chapter. She advised both chapters until retirement. A member of the ACM SIGGRAPH Digital Art committee and the ACM SIGGRAPH History Committee, Jan co-directs, along with Bonnie Mitchell, the ACM SIGGRAPH Digital Art Archive and the ACM SIGGRAPH History Archive. In addition, she co-directs the ISEA Symposium Archive with Bonnie Mitchell and Wim van der Plas.

Moderator - Valencia James. Valencia James is a Barbadian free-lance performer, maker and researcher interested in the intersection between dance, theatre, technology and activism. She believes in the power of the arts to inspire change. In 2013 Valencia co-founded the AI_am project, which explores the application of machine learning and artificial intelligence in dance. The project has been presented at several international forums such as the 2015 International Joint Conference on Artificial Intelligence in Buenos Aires and premiered their first evening-length work in Budapest and Gothenburg in 2017. Valencia also creates solo works which explore stereotypes and post-colonial narratives. She has performed extensively in Hungary, Romania, Poland, France, Israel, Sweden, Argentina, and Canada. Valencia is currently a 2020 Rapid Response for a Better Digital Future Fellow at Eyebeam Center for the Future of Journalism.